

Worksheet Problems on Data Structures and Algorithms

Worksheet 2

Q1: Find the closest pair from two sorted arrays?

Given two arrays **arr1[0...m-1]** and **arr2[0..n-1]**, and a number **x**, the task is to find the pair **arr1[i] + arr2[j]** such that absolute value of **(arr1[i] + arr2[j] –**

x) is minimum.

Input: *arr1[] = {1, 4, 5, 7};*

arr2[] = {10, 20, 30, 40};

x = 32

Output: *1 and 30*

Input: *arr1[] = {1, 4, 5, 7};*

arr2[] = {10, 20, 30, 40};

x = 50

Output: *7 and 40*

Q2: Find three closest elements from given three sorted arrays

Given three sorted arrays A[], B[] and C[], find 3 elements i, j and k from A, B and C respectively such that $\max(\text{abs}(A[i] - B[j]), \text{abs}(B[j] - C[k]), \text{abs}(C[k] - A[i]))$ is minimized. Here abs() indicates absolute value.

Input : *A[] = {1, 4, 10}*

B[] = {2, 15, 20}

C[] = {10, 12}

Output: *10 15 10*

Explanation: *10 from A, 15 from B and 10 from C*

Input: *A[] = {20, 24, 100}*

B[] = {2, 19, 22, 79, 800}

C[] = {10, 12, 23, 24, 119}

Output: *24 22 23*

Explanation: *24 from A, 22 from B and 23 from C*

Q3: Find k closest elements to a given value

Given a sorted array arr[] and a value X, find the k closest elements to X in arr[].

Input: K = 4, X = 35

arr[] = {12, 16, 22, 30, 35, 39, 42,
45, 48, 50, 53, 55, 56}

Output: 30 39 42 45

Q4: Find the element before which all the elements are smaller than it, and after which all are greater

Given an array, find an element before which all elements are smaller than it, and after which all are greater than it. Return the index of the element if there is such an element, otherwise, return -1.

Examples:

Input: arr[] = {5, 1, 4, 3, 6, 8, 10, 7, 9};

Output: 4

Explanation: All elements on left of arr[4] are smaller than it and all elements on right are greater.

Input: arr[] = {5, 1, 4, 4};

Output: -1

Explanation: No such index exists.

Q5: Find a Fixed Point (Value equal to index) in a given array

Given an array of n distinct integers sorted in ascending order, write a function that returns a Fixed Point in the array, if there is any Fixed Point present in the array, it returns -1. Fixed Point in an array is an index i such that arr[i] equals i. Note that integers in the array can be negative.

Examples:

Input: arr[] = {-10, -5, 0, 3, 7}

Output: 3

Explanation: arr[3] == 3

Input: arr[] = {0, 2, 5, 8, 17}

Output: 0

Explanation: arr[0] == 0

Input: arr[] = {-10, -5, 3, 4, 7, 9}

Output: -1

Explanation: No Fixed Point